The Temporal Geometry of Bandits With Arriving Arms

Abstract

New actions can enter a sequential experiment after substantial evidence has accumulated for existing actions. Even with stationary reward distributions, the best available action can change. We characterize the minimax regret for each deterministic arrival schedule, allowing arbitrary finite batches. The rate is a sum of learning costs over the remaining-lifetime scales, and one causal MOSS policy attains it without knowing the endpoint or future arrivals while retaining every observation. An equivalent concave allocation problem makes schedule difficulty computable. A lifetime fixed point captures the Euclidean norm of the scale costs, whereas regret depends on their sum. Under independent random arrivals, the reward environment must remain fixed across schedules, and averaging the deterministic complexities can overstate minimax risk by a logarithmic factor. For fixed batch templates with a common random time stretch, we obtain a positive characterization by a computable truncated-lifetime functional; this result assumes a known endpoint and a stretch revealed by the first arrival. Numerical studies use compatible multiscale priors, an exact small-sample Bayes benchmark, separate clock and capacity interventions, and independent evaluation of common-gap mixtures. They identify both benefits and finite-sample costs of the algorithm's design choices.

Key Words: Adaptive allocation; dynamic regret; minimax risk; sequential experimentation; sleeping bandit.

1. Introduction

1.1 Overview

An experiment may continue while new actions become available. A product comparison can admit new designs; an allocation system can receive new candidates from an external development process. Existing observations remain informative about old actions, but each arrival introduces an alternative for which the system has no data. A natural objective is to limit the cumulative loss relative to the best action available at each decision time. We study this objective when actions remain available after entry and their reward distributions do not change.

The final number of actions gives an incomplete description of this problem. Consider an incumbent and M new actions over T rounds. If all new actions are present from the beginning, the usual worst-case learning cost is of order $\min\{T, \sqrt{MT}\}$. If the same actions arrive in the last round, the regret is at most one. With several arrival batches, neither the longest remaining lifetime nor the sum of the lifetimes generally determines the rate. Learning opportunities overlap, and several time scales can each contribute appreciably to the total loss. The question is therefore which feature of the entire arrival schedule measures its statistical difficulty.

We answer this question for every deterministic exogenous schedule. Each arm has an independent, stationary reward distribution supported on $[0, 1]$. Arrivals can occur in arbitrarily large finite batches, and ties are allowed. We fix the schedule before maximizing expected regret over reward distributions. The benchmark chooses the available arm with largest mean at each round. In particular, the benchmark changes only because of arrivals, not because an existing reward distribution drifts.

The answer is a sum over remaining-lifetime scales. For a fixed endpoint, group the

competing arms whose remaining lifetimes are between $2^{\ell-1}$ and 2^{ℓ}. A group containing k_ℓ arms contributes $\min\{2^\ell, \sqrt{k_\ell 2^\ell}\}$. We show that the sum of these contributions equals the schedule-specific minimax regret up to universal multiplicative constants. A learner that knows neither the endpoint nor future arrivals achieves this rate. It uses a MOSS index with a clock that restarts when the number of competing arms doubles, while its sample counts and reward sums continue across every restart.

The main difficulty in the lower bound is to make the scale-specific testing problems coexist. Adding separate worst-case risks is invalid if their respective hard environments cannot occur together. Our construction selects a constant fraction of the relevant scale mass and places independent testing problems in disjoint time windows. The benchmark increments are summable, and the reference means do not reveal the outcomes of earlier tests. Information acquired before a testing window is included in its likelihood calculation. The upper bound faces a different issue: retaining observations prevents a conditional restart argument. We instead control all potential sample prefixes and charge each arm-count pair at most once.

A second expression makes the complexity computable without a dyadic grid. We maximize a clipped square-root objective over the fractional sample allocations allowed by the arrivals. The constraints are nested suffix capacities, and their optimizer has a lower-convex-hull representation. This is an instance of established nested resource allocation; its statistical role here is to identify a quantity equivalent to regret. The representation also gives exact stability properties, a sharp comparison with integer sample allocations and a precise explanation of diminishing marginal capacity when more arms are added.

The results distinguish several notions that are otherwise easy to conflate. A natural lifetime fixed point measures the Euclidean norm of the scale costs, while the minimax risk measures their sum. The two can differ by a factor of order $\sqrt{\log T}$. Retaining observations

can reduce regret from squared-logarithmic to logarithmic order relative to an otherwise identical policy that erases them, even though both policies have the same minimax rate.

Random arrivals introduce another distinction. The reward environment must be fixed before the schedule is drawn, so separate least-favorable environments for different schedules cannot be combined. We first exhibit a law under which the average deterministic complexity grows logarithmically while joint minimax risk stays bounded. We then characterize the joint risk for fixed batch templates whose relative times share one random stretch. A truncated-lifetime functional identifies the amount of difficulty that a common environment can sustain across stretches. The endpoint is known in this result, and the first batch reveals the remaining schedule. The constants depend on the fixed template; the result does not address growing templates or arbitrary arrival laws.

The numerical studies follow these distinctions. Compatible product priors connect the multiscale lower construction to observed regret, with an exact Bayes calculation at small horizons. A factorial comparison separates clock changes from capacity calibration, while a matched pair isolates the effect of forgetting. Independent development and evaluation samples compare one reward environment across random schedules with environments chosen separately for each schedule. The studies include failures of the proposed index and retain the information differences between the policies being compared.

1.2 Relation to Existing Work

Stochastic sleeping bandits already allow changing availability sets. Ordering arms by their means makes the best-ordering comparator coincide with the best currently available arm. The AUER analysis of Kleinberg et al. (2008) gives logarithmic bounds involving adjacent mean gaps for a finite pool and retains observations between appearances. Thus neither the comparator nor data retention is specific to the present model. Gaillard et al. (2023)

further relate sleeping internal regret to ordering regret under independent and identically distributed losses. Per-action guarantees such as those of Nguyen and Mehta (2024) use another comparator: a fixed action on the rounds when it is active. These results do not by themselves give the constant-factor minimax risk for each specified monotone arrival schedule.

The growing-action setting is studied directly by Ghalme et al. (2021). Their ballooning-bandit formulation uses the best available arm and establishes both impossibility results and sublinear guarantees under assumptions on the arrival of a best arm, with further discussion of arrival rates. Their BL-MOSS procedure uses the horizon and arrival information to select a collection of arms and subsequently stops admitting new ones. Our characterization conditions on an arbitrary fixed schedule and maximizes over all bounded stationary reward laws. It imposes no tail assumption on the arrival of an optimal arm and requires no cutoff for admitting arms.

The index itself comes from MOSS (Audibert and Bubeck 2009). Anytime minimax-optimal MOSS procedures for fixed arm sets were developed by Degenne and Perchet (2016); the coexistence of minimax and instance-dependent guarantees also has substantial precedent, including Ménard and Garivier (2017). We use this literature as the local learning component. The contribution is the schedule-specific upper and lower characterization, with an upper-bound argument that permits clock recalibration while retaining observations as the arm pool grows. Our gap bound is not an instance-optimality result.

The allocation representation is related to separable convex optimization with nested constraints. Vidal et al. (2016) explicitly derive lower-convex-hull solutions for shared-shape objectives, and Vidal et al. (2019) treat nonsmooth convex objectives within a broader class. General submodularity results for concave optimization over polymatroids are available in He et al. (2012). We use these structures with explicit proofs for the present clipped square-root objective. The connection to bandit risk, rather than a new convex optimization algorithm,

is the point of the representation.

2. Arrival Geometry and Minimax Risk

2.1 Model and Information

Arms have labels in a countable set. Fix a deterministic sequence of integer arrival times $\tau_i \geq 1$, with

$$A_t = \{i : \tau_i \leq t\}, \qquad 1 \leq K_t := |A_t| < \infty.$$

At round t, the learner observes the entire new batch before choosing $I_t \in A_t$. Arm i supplies a potential reward stream $(X_{i,n})_{n\geq 1}$ of independent and identically distributed random variables with values in $[0,1]$ and mean μ_i. Streams are independent across arms. The nth pull reveals $X_{i,n}$; all other rewards remain unobserved. Policy randomization is independent of these streams. The choice at a round may depend only on the revealed arrival sets, previous actions and observations, and this randomization.

Let $B_t = \max_{i\in A_t} \mu_i$ and $\Delta_i(t) = B_t - \mu_i$. We use the random pseudo-regret

$$R_T = \sum_{t=1}^{T} \Delta_{I_t}(t), \qquad 0 \leq R_T \leq T. \tag{1}$$

Both B_t and the gap of an already available arm are nondecreasing. No further assumption on these quantities is imposed. In particular, a new batch can contain several optimal arms or change the benchmark by any amount compatible with the reward range.

For a fixed schedule τ and deterministic endpoint T, define

$$\mathfrak{R}_T(\tau) = \inf_{\pi\in\Pi_{T,\tau}} \sup_P \mathbb{E}_P^\pi R_T. \tag{2}$$

The class $\Pi_{T,\tau}$ permits knowledge of T and of the complete arrival schedule, but not of the reward laws. The supremum ranges over the independent bounded stationary laws just described. Giving the lower-bound learner this additional information strengthens the comparison. Our upper-bound policy belongs to the smaller class that uses no future arrival information and no endpoint.

The minimax value is unchanged if the supremum is restricted to Bernoulli rewards. For the nontrivial direction, replace each observed bounded reward X by $\mathbf{1}\{U \le X\}$ with an independent uniform random variable U. The resulting stream for arm i consists of independent Bernoulli variables with mean μ_i, and the regret of the induced actions depends on the same means. Taking an arbitrarily near-optimal Bernoulli policy proves the assertion. This reduction concerns minimax risk; it does not identify the instance-specific information in a general reward law.

2.2 The Lifetime Spectrum

Choose any arm $a \in A_1$ as an anchor for counting. Its mean is unknown and it need not be optimal. Define

$$H_i = T - \tau_i + 1, \qquad L = \lceil \log_2 T \rceil, \qquad b_\ell = 2^\ell.$$

Among the arms other than a, let $G_0 = \{i : H_i = 1\}$ and

$$G_\ell = \{i : 2^{\ell-1} < H_i \le 2^\ell\}, \qquad 1 \le \ell \le L, \qquad k_\ell = |G_\ell|.$$

The scale costs and their sum are

$$s_{\ell,T} = \min\{b_\ell, \sqrt{k_\ell b_\ell}\}, \qquad \mathscr{S}_T = \sum_{\ell=0}^{L} s_{\ell,T}. \tag{3}$$

Every possible anchor has lifetime T, so the spectrum is independent of the anchor's identity. Empty groups contribute zero; if $K_T = 1$, the spectrum and regret are both zero.

Theorem 1 (Schedule-specific minimax risk)**.** *There is one causal policy π^{MC}, specified in Section 3, such that for every allowed infinite arrival schedule and every deterministic $T \geq 1$,*

$$\frac{\mathscr{S}_T}{10000} \leq \mathfrak{R}_T(\tau) \leq \sup_P \mathbb{E}_P R_T(\pi^{\mathrm{MC}}) \leq \min\{T, 300\mathscr{S}_T\}. \tag{4}$$

The policy does not use T, future arrivals, the eventual number of arms or reward means. Its sample counts and reward sums are never erased.

The constants are not optimized. Throughout, a rate characterization or the notation $\asymp$ refers to multiplicative bounds by positive universal constants, unless stated otherwise. It does not refer to an exact asymptotic minimax constant.

The truncation in $s_{\ell,T}$ matters when a batch contains more arms than the number of remaining rounds. Such a batch cannot contribute more regret than its remaining lifetime. When its size is smaller, its cost has the familiar square-root form. The theorem states that these costs accumulate across scales, despite the fact that the sampling windows overlap. Section 3 gives the upper-bound mechanism, and the supplement proves the compatible lower construction.

2.3 A Capacity Without a Grid

Write $M = K_T - 1$. A fractional allocation to the competing arms is feasible precisely when

$$\mathcal{N}_T = \left\{ n \in [0,\infty)^M : \sum_{i:H_i \leq h} n_i \leq h \text{ for } h = 1, \ldots, T \right\}. \tag{5}$$

Indeed, the arms in the displayed sum are available only in the last h time slots. Conversely, process arms from shortest to longest lifetime and place their demands in the unused part of their legal suffixes. The nested constraints ensure that the available capacity is sufficient at each step. Unused capacity can be assigned to the anchor.

Define

$$\phi(x) = \min\{x, \sqrt{x}\}, \qquad \mathscr{C}_T = \max_{n \in \mathcal{N}_T} \sum_{i \neq a} \phi(n_i), \tag{6}$$

with $\mathscr{C}_T = 0$ when $M = 0$. The feasible set is compact and ϕ is continuous, so a maximizer exists. The capacity is computed from arrival times alone. This calculation does not establish a regret guarantee for a policy based on the maximizing allocation.

Theorem 2 (Capacity characterization)**.** *For every schedule and endpoint,*

$$\tfrac{1}{4}\mathscr{S}_T \leq \mathscr{C}_T \leq \mathscr{S}_T, \qquad \frac{\mathscr{C}_T}{10000} \leq \mathfrak{R}_T(\tau) \leq \sup_P \mathbb{E}_P R_T(\pi^{\mathrm{MC}}) \leq \min\{T, 1200\mathscr{C}_T\}. \tag{7}$$

To see the upper comparison, a group has total allocated mass at most b_ℓ, and its objective is at most both that mass and the square root of k_ℓ times that mass. For the other direction, allocate $b_\ell/(4k_\ell)$ to each arm of a nonempty group. The geometric sum of group capacities verifies every suffix constraint, and each group contributes at least $s_{\ell,T}/4$. The complete argument is in the supplement.

The linear part of ϕ prevents a fractional observation from contributing more than its mass. Replacing ϕ by an untruncated square root would assign unbounded capacity to a sufficiently large batch arriving in the last round. In contrast, (6) gives exactly one for every nonempty such batch.

3. A Causal Policy That Retains Observations

3.1 Clock Changes and Lifetime Statistics

For each available arm, store its lifetime sample count N_i and reward sum S_i. Set an integer variable base to zero initially. If only one arm is available, pull it and update its statistics. Otherwise, after receiving the current batch, start a new clock segment whenever

$$\text{base} = 0 \quad \text{or} \quad K_t - 1 \geq 2\,\text{base}.$$

At that time set

$$\text{base} = K_t - 1, \qquad \sigma = t, \qquad d = 2\,\text{base}. \tag{8}$$

At all other times retain σ and d. The local clock is $s = t - \sigma + 1$. Select an arm maximizing

$$U_i(t) = \begin{cases} +\infty, & N_i(t-1) = 0, \\ \widehat{\mu}_{i,N_i(t-1)} + \sqrt{\dfrac{[\log\{s/(dN_i(t-1))\}]_+}{N_i(t-1)}}, & N_i(t-1) \geq 1, \end{cases} \tag{9}$$

where $\widehat{\mu}_{i,n}$ is the average of the first n rewards from arm i. Break ties by arrival order and a fixed ordering within a batch. Observe one reward, then increment the chosen arm's lifetime count and sum. No clock change modifies either statistic. We call this policy MC-MOSS, for MOSS with a recalibrated clock and retained memory.

Every decision is well defined even if a batch is too large to initialize fully. A batch that crosses a threshold belongs to the new segment, since (8) precedes the action. If the segment begins with c competing arms, its total number of available arms stays at most $d = 2c$: otherwise a new segment would have begun before the next action. This capacity bound holds without predicting the final arm count.

3.2 Why Old Observations Can Be Used

The policy does not restart a statistical experiment at a clock change. Conditional on the observed past, old empirical means need not have the distribution of fresh sample averages. We therefore work on the original potential reward streams and take unconditional expectations throughout the local argument.

For one stream of mean μ, let

$$Z(v) = \sup_{n \geq 1} \left[\mu - \widehat{\mu}_n - \sqrt{[\log(v/n)]_+ / n} \right]_+ . \tag{10}$$

A maximal concentration argument gives

$$\mathbb{E}Z(v) \leq 5v^{-1/2}, \qquad \mathbb{P}\{Z(v) > x\} \leq \min\{1, 5/(vx^2)\}. \tag{11}$$

At each physical round, choose a deterministic label for a best available arm, using the fixed means and schedule. On every action selecting an observed arm, index comparison yields

$$\Delta_{I_t}(t) \leq Z_{i_t^*}(s/d) + (\widehat{\mu}_{I_t,n} - \mu_{I_t})_+ + \sqrt{[\log\{s/(dn)\}]_+ / n}, \qquad n = N_{I_t}(t-1). \tag{12}$$

If the best arm is still unobserved, the infinite-index rule selects an unobserved arm, which is charged separately.

For the other terms, each pair consisting of an arm and its count just before a pull occurs at most once. If at most h such pulls use k arms, this pathwise fact gives

$$\mathbb{E}\sum (\widehat{\mu}_{I_t,n} - \mu_{I_t})_+ \leq \sqrt{2kh}, \qquad \sum_{n \geq 1} \sqrt{[\log(v/n)]_+ / n} \leq \sqrt{2\pi v}.$$

No independence between a random count and its empirical mean is used. Old samples may

cause some counts to be skipped, which can only reduce these upper sums.

Lemma 1 (A segment with retained history)**.** *Consider a segment with deterministic starting time and capacity d, using* (9) *and all rewards gathered in the experiment. For any deterministic prefix of length h within the segment, its expected regret satisfies*

$$\mathbb{E}R_{\text{segment}} \leq \min\{h, 15\sqrt{dh}\}. \tag{13}$$

Arrivals within the segment are allowed, provided at most d arms are available. The expectation is under the original experiment, not conditional on an arbitrary realized history.

The terms above sum to at most $(1 + 10 + \sqrt{2} + \sqrt{2\pi})\sqrt{dh}$. The extra one accounts for genuine first observations. To pass from segments to (4), note that their initial competing-arm counts grow geometrically. Group the segment starts by remaining-lifetime scale, and assign each segment to the arms that caused its capacity increase. Delaying these arms to the segment start changes the lifetime spectrum by at most a factor of two. This charges the sum of the local square-root costs to $\mathscr{S}_T$. Once the competing-arm count exceeds the remaining horizon, the entire tail is charged by its length. Full details and the lower-bound construction appear in the supplement.

3.3 Instance Behavior and the Role of Memory

For each arm that is strictly suboptimal at some time before T, define

$$\gamma_i = \min_{t \leq T:\, i \in A_t,\, \Delta_i(t) > 0} \Delta_i(t), \qquad \gamma = \min_i \gamma_i.$$

These positive quantities are defined retrospectively and are not inputs to the policy. They need not be bounded away from zero as the experiment grows.

Proposition 1 (Gap bounds and a paired restart comparison)**.** *If at least one positive gap occurs, MC-MOSS satisfies*

$$\mathbb{E}R_T \leq \min\left\{T, 300\mathscr{S}_T, \frac{400(K_T-1)\log(eT)}{\gamma}\right\}. \tag{14}$$

For deterministic rewards, the stronger pathwise bound

$$R_T \leq \sum_{i:\gamma_i \text{ defined}} \left\{1 + \frac{2\log(eT)}{\gamma_i}\right\} \tag{15}$$

holds. There is a deterministic fixed-gap schedule family on which

$$R_T(\pi^{\mathrm{MC}}) = O(\log T), \qquad R_T(\pi^{\mathrm{cold}}) = \Omega\{(\log T)^2\},$$

where π^{cold} has the same indices, clock changes and tie rules, but erases all statistics at each clock change. Both policies obey the upper rate in Theorem 1.

The separating family has one inferior old arm and newly arriving arms tied with the incumbent. Erasing observations repeatedly removes the evidence that the old arm is inferior. The proof counts actual inferior-arm pulls; it does not compare two regret upper bounds. This comparison isolates a cost of forgetting. It makes no assertion of dominance over AUER, ordinary all-history UCB, or other MOSS variants.

A simple boundary example further separates minimax and instance behavior. Suppose the initial arm always gives reward one, and one arm giving reward zero arrives at every later round. The infinite-index rule explores every new arm and incurs regret $T-1$. Capping observed indices at one and assigning index one to unobserved arms, with the same tie order, makes regret zero after the initial observation. The capped policy retains the bounds in Theorem 1 and Proposition 1; its proof is given in the supplement. The worst-case risk for

this arrival schedule is linear, so both behaviors are compatible with minimax optimality.

4. Accumulation Across Lifetime Scales

4.1 What a Lifetime Fixed Point Retains

A useful scalar summary discards arms with at most r rounds left and compares the total lifetime of the remaining arms with r^2. For $M \geq 1$, define

$$V_T(r) = \sum_{i \neq a} H_i \mathbf{1}\{H_i > r\}, \qquad \Gamma_T = \min\{r \in \{1, \ldots, T\} : V_T(r) \leq r^2\}. \tag{16}$$

Set $\Gamma_T = 0$ for $M = 0$. This functional is introduced here as a candidate compression of the arrival information. Its relationship to the exact rate can be characterized for every schedule.

Theorem 3 (A squared-spectrum interpretation)**.** *Let* $Q_T = (\sum_\ell s_{\ell,T}^2)^{1/2}$. *When* $M \geq 1$,

$$\Gamma_T \leq \lceil Q_T \rceil \leq 2Q_T, \qquad Q_T \leq 3\Gamma_T. \tag{17}$$

Consequently, with

$$D_T^{\mathrm{eff}} = \frac{\mathscr{S}_T^2}{Q_T^2}, \tag{18}$$

we have $1 \leq D_T^{\mathrm{eff}} \leq 1 + \lceil \log_2 T \rceil$ *and*

$$\mathfrak{R}_T(\tau) \asymp \mathscr{S}_T \asymp \Gamma_T \sqrt{D_T^{\mathrm{eff}}}.$$

Moreover, $\mathscr{S}_T \leq C\Gamma_T \sqrt{1 + \log(T/\Gamma_T)}$ *for a universal* C.

The proof separates scales below and above Γ_T. The small scales have a bounded geometric square sum. Above the threshold, their squared costs are bounded by twice $V_T(\Gamma_T)$.

In the other direction, any scale above Q_T must be unsaturated, so its squared cost contains the corresponding lifetime mass. This yields (17) without assumptions on the spacing or sizes of batches.

The effective number D_T^{eff} is large when many scales contribute comparable amounts. If a bounded number of scales dominate, the fixed point already determines the rate up to constants. A growing discrepancy requires contributions to accumulate across an increasing number of scales. Although D_T^{eff} uses the specified dyadic partition, the capacity and the fixed point themselves do not.

4.2 A Sharp Separation

Let $T = 16^m$. In addition to the anchor, batch j contains 4^j arms with common remaining lifetime $H_j = 4^{2m-j}$, for $j = 0, \dots, m$. These lifetimes occupy different dyadic groups and satisfy $4^j H_j = T$. Hence

$$\mathscr{S}_T = (m+1)\sqrt{T}, \qquad Q_T = \sqrt{(m+1)T}, \qquad \mathfrak{R}_T(\tau) \asymp \sqrt{T}\log T. \tag{19}$$

The fixed point is of order $\sqrt{T \log T}$ and the final number of arms is of order $\sqrt{T}$. The usual total-count expression $\sqrt{K_T T}$ is then of order $T^{3/4}$, while the fixed point is too small by a factor of order $\sqrt{\log T}$. The final bound in Theorem 3 shows that the latter loss has the largest possible order.

Even the pair (K_T, Γ_T) is insufficient at constant precision. At the same endpoint, keep the same total number of arms, put $m+1$ competitors at time one, and put all remaining competitors in the last round. This schedule has $\Gamma_T \asymp \sqrt{mT}$ as before, but $\mathscr{S}_T \asymp \sqrt{mT}$. The two schedules therefore agree in final count and in the order of their fixed points, while their minimax risks have an unbounded ratio.

4.3 The Boundary for Learning

Theorem 4 (Uniform sublinear regret)**.** *For one infinite deterministic arrival schedule, the following are equivalent:*

$$K_t = o(t) \iff \mathscr{S}_T = o(T) \iff \mathscr{C}_T = o(T)$$
$$\iff \text{there is one policy } \pi \text{ with } \sup_P \mathbb{E}_P R_T(\pi) = o(T).$$

The same MC-MOSS policy gives the last property for every schedule satisfying the first. These conditions are also equivalent to uniform convergence of R_T/T to zero in probability under a single policy.

For sufficiency, Cauchy–Schwarz gives $\mathscr{S}_T \le 2\sqrt{(K_T-1)T}$. For necessity, if $K_t \ge \epsilon t$ along a subsequence, evaluate regret at $T = 2t$. Those already present competitors all have lifetime at least $t+1$, forcing a linear spectral cost. The uniform supremum is essential: the theorem does not assert that a single fixed reward environment is hard at every endpoint.

The characterization gives familiar rates when a fixed fraction of the final competitors arrive by $T/2$. In that case,

$$\mathfrak{R}_T(\tau) \asymp \min\{T, \sqrt{(K_T-1)T}\},$$

with constants depending on that fraction. Thus $K_t \asymp t^\alpha$, $0 < \alpha < 1$, gives rate $T^{(1+\alpha)/2}$; $K_t \asymp t/(\log t)^p$, $p > 0$, gives $T/(\log T)^{p/2}$; and $K_t \asymp t/\log\log t$ gives $T/\sqrt{\log\log T}$. These are order statements for the specified growth conditions. They show why an extra concentration logarithm in a particular upper bound must not be interpreted as a stronger learnability requirement.

5. Computing and Interpreting Arrival Capacity

5.1 A Hull Calculation and Its Certificate

Sort the M competing-arm lifetimes as $H_{(1)} \le \cdots \le H_{(M)}$. The suffix constraints are equivalent to

$$\sum_{i=1}^{r} n_{(i)} \le H_{(r)}, \qquad r = 1, \ldots, M.$$

For tied lifetimes the constraint at the last tied index implies the earlier ones. Let $0 = r_0 < r_1 < \cdots < r_B = M$ be the vertices of the lower convex hull of $(0,0), (r, H_{(r)})$, with $H_{(0)} = 0$. Its successive slopes are

$$q_b = \frac{H_{(r_b)} - H_{(r_{b-1})}}{r_b - r_{b-1}}, \qquad b = 1, \ldots, B. \tag{20}$$

They are positive and nondecreasing.

Proposition 2 (Explicit capacity and dual certificate)**.** *Assign $n^*_{(i)} = q_b$ for $r_{b-1} < i \le r_b$. This allocation maximizes* (6)*, so*

$$\mathscr{C}_T = \sum_{b=1}^{B} (r_b - r_{b-1}) \phi(q_b). \tag{21}$$

An exact dual representation is

$$\mathscr{C}_T = \min_{\lambda_r \ge 0} \left\{ \sum_{r=1}^{M} H_{(r)} \lambda_r + \sum_{i=1}^{M} g\left(\sum_{r=i}^{M} \lambda_r \right) \right\}, \tag{22}$$

where $g(0) = +\infty$ *and*

$$g(c) = \begin{cases} 1/(4c), & 0 < c \le 1/2, \\ 1 - c, & 1/2 \le c \le 1, \\ 0, & c \ge 1. \end{cases}$$

The minimum is attained by nonzero multipliers only at hull vertices.

This is the common-shape nested allocation geometry studied by Vidal et al. (2016), with a supergradient treatment of the corner in ϕ. If $p_b = 1$ for $q_b \le 1$ and $p_b = 1/(2\sqrt{q_b})$ otherwise, an attaining certificate is $\lambda_{r_b} = p_b - p_{b+1}$, where $p_{B+1} = 0$, and zero elsewhere. Sorting takes $O(M \log M)$ comparisons; a monotone stack then builds the hull in $O(M)$ arithmetic operations. Integer cross-products can determine the slope order without floating-point comparisons. No claim about unit-cost arithmetic for arbitrarily large integers is needed.

5.2 Integer Allocations

Restricting sample allocations to integers changes the capacity by only a small multiplicative factor. To quantify this difference, define the integer allocation capacity

$$\mathscr{C}_T^{\text{int}} = \max_{N \in \mathcal{N}_T \cap \mathbb{Z}_+^M} \sum_i \sqrt{N_i}.$$

An integer feasible vector can be realized by actual pulls, again filling legal suffixes from shortest to longest lifetime.

Theorem 5 (Sharp comparison with integer capacity)**.** *For every finite arrival schedule,*

$$\kappa \mathscr{C}_T \le \mathscr{C}_T^{\text{int}} \le \mathscr{C}_T, \qquad \kappa = \frac{2\,2^{1/4}}{1+\sqrt{2}} = 0.985171\ldots. \tag{23}$$

No larger universal constant can replace κ *in the lower bound.*

The upper comparison follows because $\phi(N_i) = \sqrt{N_i}$ at integer arguments. For the lower comparison, sort a fractional optimizer and round all of its prefix sums with the same uniform offset. The resulting integer prefix sums remain feasible, every coordinate is rounded to one of its two nearest integers, and its expectation is unchanged. The linear interpolation of the square root between integer arguments is at least $\kappa\phi$. Balanced allocations with average count tending to $\sqrt{2}$ attain this loss asymptotically. The full rounding and sharpness arguments are in the supplement. The fractional and integer capacities are therefore close; this comparison does not supply a regret guarantee for a policy based on either optimizing allocation.

5.3 Sensitivity and Marginal Contributions

The capacity has several exact properties, which need not transfer as exact inequalities to the minimax risk. If a batch of any finite size is added with every lifetime at most h, the capacity increases by at most h. If two lifetime vectors on the same arm set obey $\max_i |H_i - H'_i| \le d$, then

$$|\mathscr{C}(H) - \mathscr{C}(H')| \le d. \tag{24}$$

Along one infinite schedule, even with an arbitrarily large new batch,

$$0 \le \mathscr{C}_{T+1} - \mathscr{C}_T \le 1. \tag{25}$$

These facts follow by deleting at most h, d, or one unit of allocated mass and using that ϕ is increasing and 1-Lipschitz. Their complete proofs handle tied lifetimes and are given in the supplement.

A more detailed description fixes a finite universe of possible competing arms and varies its subset A. Let $\mathscr{C}(A)$ denote the corresponding capacity, with the anchor always present.

For $z > 0$, define

$$F_z(A) = \min_{0 \le h \le T,\ h \in \mathbb{Z}} \{h + z|\{i \in A : H_i > h\}|\}. \tag{26}$$

This is the largest total feasible allocation if each arm's contribution is capped at z. The hull allocation attains every capped objective simultaneously, and

$$\mathscr{C}(A) = \tfrac{1}{2}F_1(A) + \tfrac{1}{4}\int_1^\infty F_z(A) z^{-3/2}\, dz. \tag{27}$$

Each capped value is submodular. Thus, if $A \subseteq B$ and $i \notin B$,

$$0 \le \mathscr{C}(B \cup \{i\}) - \mathscr{C}(B) \le \mathscr{C}(A \cup \{i\}) - \mathscr{C}(A) \le \sqrt{H_i}.$$

This specializes the concave-polymatroid value property of He et al. (2012); the explicit cuts describe it in terms of the arrival times. It states that the marginal increase in this measure of difficulty diminishes as the candidate pool grows. Removing an available arm changes the decision benchmark, so this property is not a justification for excluding arms while claiming the original regret guarantee.

6. Independent Random Arrivals and the Order of Minimax Operations

6.1 Why Deterministic Risks Cannot Simply Be Averaged

The deterministic analysis immediately supplies an upper bound for a random schedule independent of the reward streams and policy randomization. Conditional on its full realization,

the potential streams keep their original laws. Hence, for each fixed environment,

$$\mathbb{E}R_T(\pi^{\mathrm{MC}}) \le \mathbb{E}_\tau \min\{T, 300\mathscr{S}_T(\tau)\} \le \min\{T, 1200\mathbb{E}_\tau\mathscr{C}_T(\tau)\}. \tag{28}$$

There is no corresponding general lower bound obtained by averaging Theorem 1. Its least favorable reward environment may depend on the realized schedule. When the environment must be chosen before an independent schedule draw, these choices are no longer admissible. Random-horizon minimax problems have been studied in online learning: for a restricted adversarial full-information loss game, Luo and Schapire (2014) show that the minimax value does equal the average of the fixed-horizon values. The stationary bandit construction below has a different conclusion and different information assumptions.

To formalize the distinction, let $\mathcal{Q}$ be a distribution over arrival schedules and put

$$\mathfrak{R}_T^{\mathrm{joint}}(\mathcal{Q}) = \inf_\pi \sup_P \mathbb{E}^\pi_{\tau\sim\mathcal{Q},P} R_T.$$

Here P is fixed independently of τ. The policy may know T and $\mathcal{Q}$, and sees arrivals causally. This information class is specified separately from that of MC-MOSS; the result below establishes a boundary of the deterministic-schedule characterization.

Theorem 6 (A logarithmic separation)**.** *For every integer $m \ge 1$, let $T_m = 4^m$. There is an independent two-arm arrival law $\mathcal{Q}_m$ such that*

$$\mathbb{E}_\tau\mathscr{C}_{T_m}(\tau) = c_m(m+1), \quad \frac{1}{16} \le \mathfrak{R}^{\mathrm{joint}}_{T_m}(\mathcal{Q}_m) \le 380, \quad c_m = \left(\sum_{j=0}^{m} 2^{-j}\right)^{-1}. \tag{29}$$

Thus $\mathbb{E}_\tau\mathfrak{R}_{T_m}(\tau)$ can be of order $\log T_m$, while the joint minimax risk is of order one.

The anchor arrives at time one. Draw J with probability $c_m 2^{-j}$ on $j = 0, \dots, m$, and let the only competitor have remaining lifetime $H = 4^J$. Its conditional capacity is $\sqrt{H}$, giving

the first equality. If the two means have fixed gap $\Delta > 0$, a known-horizon two-arm MOSS policy, started when the competitor arrives, satisfies

$$\mathbb{E}R_H \le \min\left\{H\Delta, \frac{50[1+\log_+(H\Delta^2)]}{\Delta}\right\}.$$

The known endpoint reveals H at arrival, so this construction is causal. Averaging its bound gives a sum over $x_j = \Delta 2^j$ of terms bounded by

$$\min\{x_j, 50(1+2\log_+ x_j)/x_j\}.$$

The terms below one have a bounded geometric sum. The terms above one also sum to a constant, because their denominator doubles while the logarithm increases linearly. A two-point testing argument on the event $H = 1$ gives the lower constant. The supplement proves the local MOSS bound and both claims in (29).

Every remaining-lifetime scale can be hard when it is allowed its own gap. A single gap chosen before the random schedule is drawn cannot be simultaneously worst at all these scales. This is the source of the separation. It leaves the conditional upper bound (28) intact, but rules out its general interpretation as a matching characterization of the independent-random-arrival minimax problem.

6.2 A Positive Result for a Revealing Batch Template

The obstruction has a computable resolution when the identities and relative times of the batches are fixed and only their common time scale is random. Fix disjoint nonempty batches $B_1, \dots, B_b$ and $\alpha_1 > \cdots > \alpha_b = 1$. Draw a positive integer H from a known finite law Q, with

$\alpha_j H$ integer for every batch. For a known endpoint $T \geq \alpha_1 \max \operatorname{supp}(Q)$, batch j arrives at

$$\tau_j = T - \alpha_j H + 1.$$

The anchor alone is available before the first batch. Labels and reward laws are fixed before H is drawn. The first arrival reveals $H = (T - \tau_1 + 1)/\alpha_1$; using this information is permitted by the causal observation model.

Define

$$W(Q) = \sup_{x \geq 1} \frac{\mathbb{E}_Q \min\{H, x\}}{\sqrt{x}} = \max_{h \in \operatorname{supp}(Q)} \frac{\mathbb{E}_Q \min\{H, h\}}{\sqrt{h}}. \tag{30}$$

For a chosen testing scale x, the numerator counts the expected number of rounds in which that test can be carried out, truncated at x. Division by $\sqrt{x}$ reflects the gap scale that keeps those observations difficult. The same x is used across every realization of the arrival process.

Theorem 7 (Fixed-template random arrivals)**.** *For the preceding experiment,*

$$\frac{W(Q)}{32} \leq \mathfrak{R}_T^{\text{joint}} \leq A_{\boldsymbol{\alpha},\boldsymbol{B}} W(Q), \tag{31}$$

where, writing $k_j = 1 + \sum_{r \leq j} |B_r|$, $d_j = \alpha_j - \alpha_{j+1}$ *for* $j < b$, *and* $d_b = 1$, *one may take*

$$A_{\boldsymbol{\alpha},\boldsymbol{B}} = \sum_{j=1}^{b} (20k_j + 43)(k_j - 1)\sqrt{d_j}.$$

The upper bound is attained by a causal rule that plays the anchor until the first batch, then starts fixed-horizon MOSS with fresh statistics at each batch, using the revealed interval length $d_j H$. *The lower bound allows the learner to use all observations and even to know* H *from the beginning.*

The constants are uniform in Q, its support range and T, with the batch template held

fixed. They are not uniform in a growing number of arms or batches. The attaining rule is a known-endpoint construction; this theorem does not assign the same guarantee to MC-MOSS or imply that cold restarting is preferable in general.

The lower bound uses a single final-batch candidate with mean $1/2 \pm 1/(8\sqrt{x})$ and sets every other mean to $1/2$. Earlier observations carry no information about the sign. Testing in the first $\min\{H, x\}$ rounds after that candidate arrives yields the numerator in (30), with a prior independent of H. For the upper bound, the positive gap of a fixed arm within a fixed batch interval is the same for every stretch. A tail integration argument bounds the average of its fixed-horizon MOSS cost by a constant times $W(Q)$. Summing the interval costs includes all rounds after the first arrival. The supplement proves the constants and justifies the use of fresh future reward streams after a reset.

The functional has a simple tail interpretation:

$$V(Q) \le W(Q) \le 2V(Q), \qquad V(Q) = \max_{h \in \mathrm{supp}(Q)} \sqrt{h}\, \mathbb{P}_Q(H \ge h). \tag{32}$$

This is the usual weak tail quantity of $\sqrt{H}$, used here as a statistical complexity. It is computable from the arrival law alone. For the fixed template, deterministic capacity is between $\sqrt{H}$ and $\sqrt{(k_b - 1)\alpha_1 H}$. Consequently,

$$\frac{\mathbb{E}_Q \mathscr{C}_T(\tau(H))}{\mathfrak{R}_T^{\mathrm{joint}}} \asymp_{\boldsymbol{\alpha}, \boldsymbol{B}} D(Q), \qquad D(Q) = \frac{\mathbb{E}_Q \sqrt{H}}{W(Q)} \ge 1. \tag{33}$$

Thus averaging the deterministic capacity remains correct, up to fixed-template constants, precisely for families of laws with bounded $D(Q)$. A deterministic scale gives $D = 1$, and support in $[h_0, Lh_0]$ gives $D \le \sqrt{L}$. For $\mathbb{P}(H = 4^j) = c_m 2^{-j}$, $W(Q) < 3$ while $\mathbb{E}\sqrt{H} = c_m(m+1)$. For a noninteger template, use $H = q4^J$ with a fixed positive integer q making every $q\alpha_j$ an integer; both quantities are then multiplied by $\sqrt{q}$. The earlier

separation therefore extends to every fixed multi-batch template in this class. Unknown endpoints, nonrevealing arrival laws and growing templates remain separate problems.

7. Numerical Studies of the Statistical Mechanisms

The studies examine three distinct questions: how compatible learning problems accumulate across lifetime scales, how the exploration clock and capacity calibration affect finite-sample behavior, and how a common reward environment interacts with a random schedule. All reported losses are pseudo-regret relative to the best currently available mean. The data are generated from the stated Bernoulli models. Monte Carlo standard errors describe simulation uncertainty; they are not uncertainty about a minimax constant.

7.1 Risk on a Compatible Multiscale Family

The first study evaluates the common null/spike family used in the lower-bound proof. At endpoint $T \in \{1024, 4096, 16384, 65536\}$, batch j has $2 \cdot 4^j$ competitors and lifetime $T/4^j$, for $j = 0, \ldots, L-1$. We use $L \leq 3$ at the two smaller endpoints and $L \leq 4$ at the two larger ones. All groups survive the compatibility selection and contribute equally to the spectrum: $\mathscr{S}_T = L\sqrt{2T}$ and $D_T^{\text{eff}} = L$.

Each group has one reference arm. With probability $1/2$ all its candidates are below the reference; otherwise one uniformly selected candidate is above it. Independent configurations and deterministic increments in the reference means give one compatible stationary environment per replication. We use the proof's gap constant and the prespecified multipliers $a = 4, 8$, with $a = 4$ in the main comparison. Larger multipliers preserve a legal prior but need not preserve the proof's total-variation bound. Each cell averages 128 independent draws of the environment and its Bernoulli observations. These finite-prior risks do not estimate the full minimax value.

For each multiscale schedule we construct a second schedule with the same endpoint and final number of arms: $2L$ competitors arrive at time one and the rest in the last round. The two fixed points differ by at most a factor 1.235 in the simulated cases. At $T = 65536$ and $L = 4$, both equal 725, whereas the spectra are 1448.15 and 725.08. Under each schedule's native prior, MC-MOSS has mean regret 146.67 (Monte Carlo standard error 3.00) on the multiscale schedule and 87.57 (3.31) on the compressed schedule.

Because that comparison changes both schedule and prior, we also run each sampled mean vector on both schedules, using the same calendar reward uniforms. Compression reduces MC-MOSS regret by 18.82 (paired standard error 5.84) for the multiscale prior and by 185.04 (3.52) for the compressed prior. The difference separates the effect of arrival timing from the choice of a difficult environment.

Figure 1 reports the normalized losses and matched comparisons. At each fixed L, normalization by $\mathscr{S}_T$ brings the endpoint curves close together, although the normalized risk still varies with L. Four effective scales do not demonstrate the asymptotic separation. In the largest four-scale case, the disjoint testing windows contribute 42.59 units of mean MC-MOSS regret, with 104.07 outside their union; the full loss enters every comparison. Cold MC improves on MC-MOSS under this native prior by 5.92 (paired standard error 1.38), but loses 40.44 (2.00) for the compressed prior run on the multiscale schedule. The finite-sample effect of retaining history depends on the environment.

For a six-environment prior at $T \in \{4, 6, 8, 10\}$, exact dynamic programming gives a separate Bayes reference with known prior and schedule. At $T = 8$, with gap increments $1/16$ and $1/8$, its risk is 0.289612, compared with 0.368656 for MC-MOSS and 0.431156 for Cold MC. The optimum is a certified lower bound for deterministic-schedule minimax risk, with more information than the anytime policies; the prior need not be least favorable. The computational supplement gives the recursion and its proof, all eight exact calculations, gap

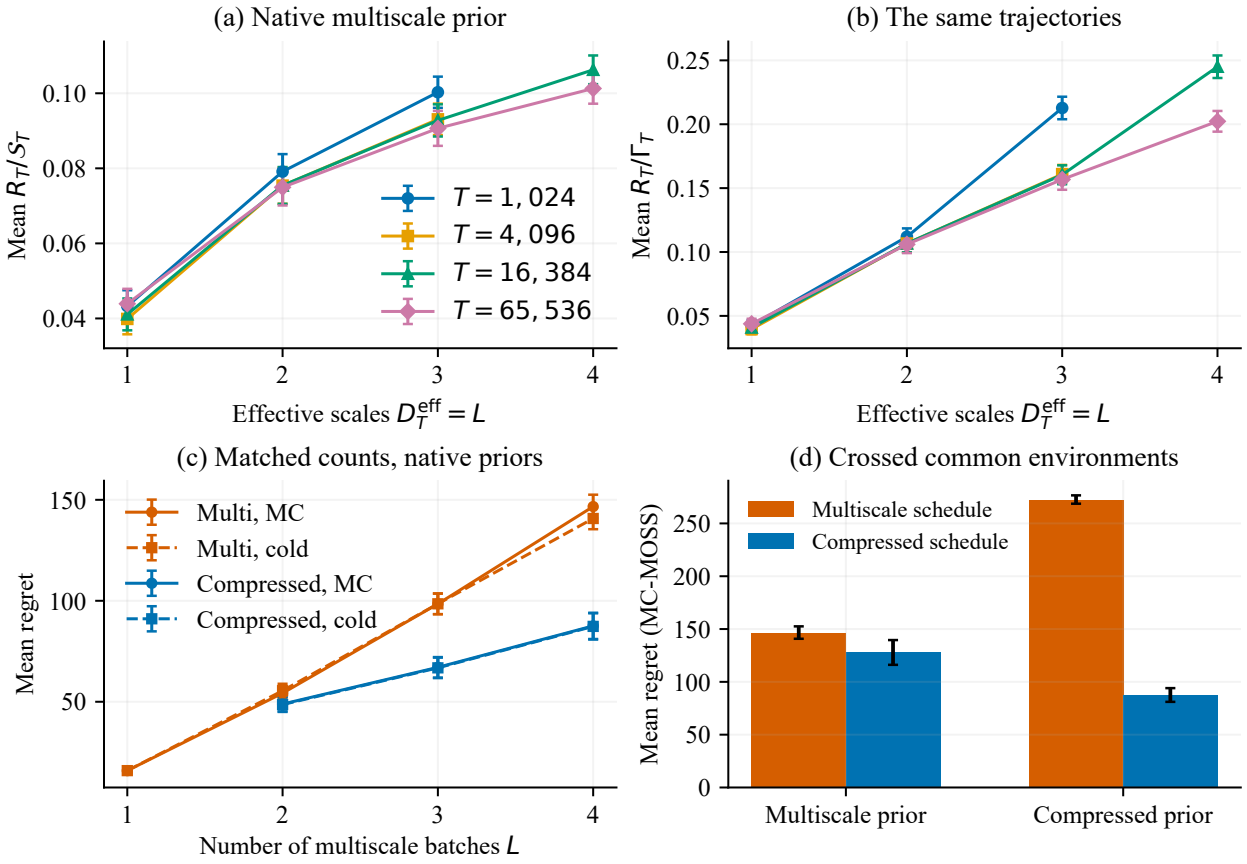


Figure 1. Bandit risk on a compatible finite product prior. Panels (a) and (b) use MC-MOSS and the native multiscale prior with gap multiplier four. Panel (c) compares each schedule under its native prior at $T = 65536$; paired schedules have the same final count. Panel (d) fixes $T = 65536$, $L = 4$, and the sampled mean vector while changing its arrival schedule. Bars are 1.96 Monte Carlo standard errors over 128 independent replications.

sensitivity, and individual window losses.

7.2 Separating Clock Calibration, Capacity, and History

The next experiment separates the two changes hidden in a comparison with a global-clock index. We cross the clock, s_t or t, with the capacity, d_j or $2(K_t - 1)$, while retaining all observations in each of the four rules. Every rule uses the index $\widehat{\mu}_i + \{\log_+(u_t/(v_t N_i))/N_i\}^{1/2}$,

with the same initialization and tie order. MC-MOSS is the local-clock, frozen-capacity cell. A fifth rule uses precisely that cell but discards its observations at the same capacity-doubling times. These contrasts isolate finite-sample effects; no spectral guarantee is asserted for the other three cells.

We also compare UCB1, AUER, Bernoulli KL-UCB, Thompson sampling, and clipped MC-MOSS. UCB1 and AUER use respective bonuses $\{2\log(t)/N_i\}^{1/2}$ and $\{8\log(t)/N_i\}^{1/2}$ (Auer et al. 2002; Kleinberg et al. 2008). KL-UCB uses the Bernoulli divergence and the original paper's recommended experimental choice $c = 0$ (Garivier and Cappé 2011); Thompson sampling starts each arriving arm with a Beta$(1, 1)$ prior (Agrawal and Goyal 2012). These are full-history arriving-arm implementations of the stated rules. We do not transfer their fixed-arm guarantees to the present arrival model.

There are 32 candidates in batches of sizes $4, 4, 8, 8, 8$, arriving at $1, \lfloor T/8\rfloor + 1, \lfloor T/3\rfloor + 1, \lfloor T/2\rfloor + 1, \lfloor 3T/4\rfloor + 1$, besides the initial anchor. We fix five mean vectors: near ties, poor candidates, a persistent new record at the fourth batch, a mixed-quality final batch, and an anchor at the reward upper bound. The full vectors are given in the supplement. Both $T = 2048$ and $T = 8192$ are used. Each cell has 128 independent replications, coupled across methods by reward uniforms at each calendar round. We repeat the entire design with the order of labels reversed within every candidate batch; the anchor remains first. Parameters were fixed before simulation, and software development used separate seeds.

Figure 2 shows why the two factors must be separated. At $T = 8192$, with forward labels and a mixed final batch, the global clock costs 26.38 (paired standard error 7.34) relative to the local clock when capacity is frozen. With the global clock held fixed, using current capacity reduces regret by 26.57 (1.30). The combined global/current rule consequently has mean regret 224.24, almost the same as MC-MOSS's 224.42, although the component contrasts are substantial. Current capacity also improves on frozen capacity under the local

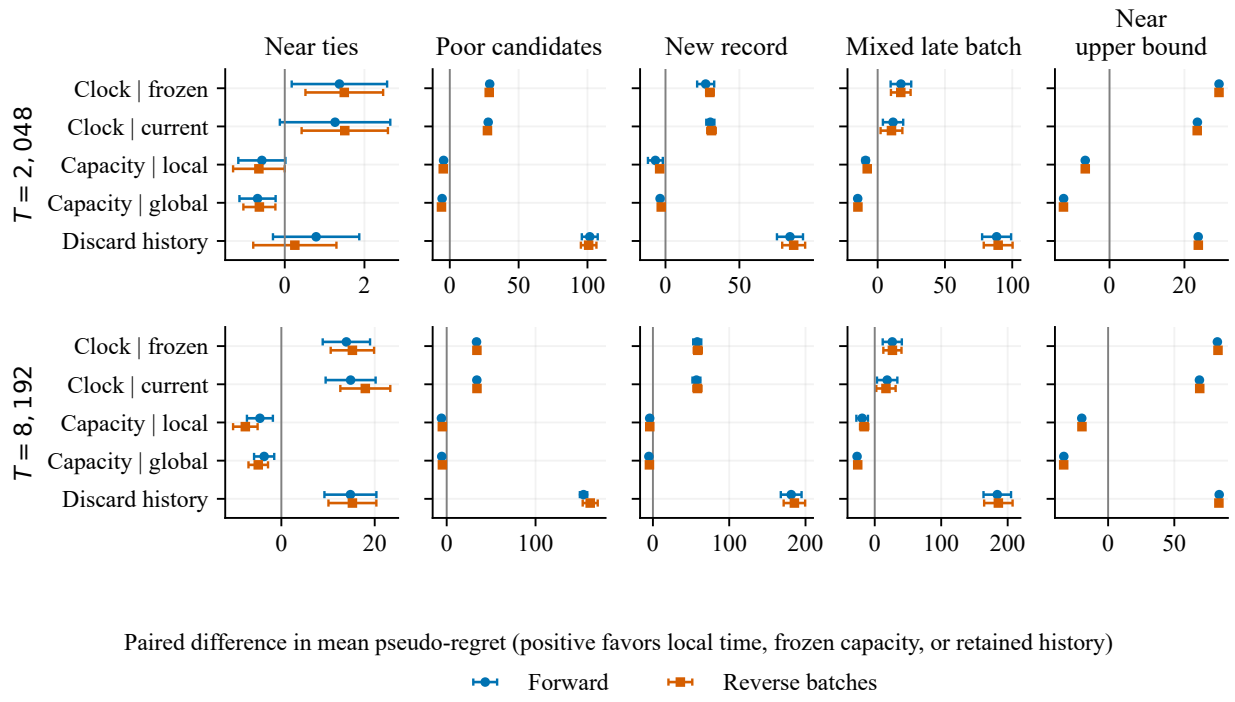


Figure 2. Matched mechanism contrasts under two horizons and two batch label orders. Clock contrasts are global minus local at fixed capacity; capacity contrasts are current minus frozen at fixed clock. Discarding history is compared with MC-MOSS at identical reset times. Points are paired mean differences and bars are 1.96 Monte Carlo standard errors from 128 independent replications. The intervals are pointwise, without adjustment for the multiple displayed comparisons. Horizontal scales differ across panels.

clock, by 18.90 (4.45) in this example. Thus the frozen capacity is a device that supports the uniform analysis, not a claim of the best finite-sample calibration. Changing the clock to global increases mean regret in every displayed cell, whereas discarding history is especially costly when inferior candidates recur. For poor candidates at $T = 8192$, the latter cost is 154.04 (2.12). Reversing labels preserves these broad patterns; the near-tie differences at $T = 2048$ are comparatively small and uncertain.

The stronger baselines also identify a substantial failure of the unclipped rule (Table 1).

Table 1. Pseudo-regret for four policies under five reward configurations.

Configuration	MC-MOSS	KL-UCB	Thompson	Clipped MC
Near ties	103.85 (2.89)	132.56 (0.50)	125.81 (0.86)	104.16 (2.69)
Poor candidates	146.44 (0.86)	236.00 (1.33)	170.35 (1.12)	147.36 (0.80)
New record	157.11 (6.02)	371.66 (3.06)	242.72 (3.82)	147.84 (4.55)
Mixed late batch	224.42 (6.91)	511.85 (5.57)	304.10 (3.43)	230.88 (7.33)
Near upper bound	116.78 (0.36)	1.70 (0.00)	8.60 (0.30)	0.00 (0.00)

Notes: Entries are mean pseudo-regret with Monte Carlo standard errors in parentheses at $T = 8192$, forward labels, and 128 replications. The complete two-horizon, two-label-order results, including UCB1 and AUER, are in the computational supplement and released CSV files.

With anchor mean one and candidate means 0.94 or 0.985, MC-MOSS incurs 116.78 mean regret at $T = 8192$, compared with 1.70 for KL-UCB and 8.60 for Thompson sampling. Clipped MC-MOSS has zero regret here: its indices never exceed one and the specified tie rule keeps selecting the anchor. This conclusion uses the exact boundary mean and the anchor's priority; it is not an assertion of uniform dominance by clipping. Indeed, in the mixed-final-batch configuration clipping changes regret by 6.46 (8.76), with no resolved improvement. The comparisons support the roles of clock calibration and history reuse while showing that a minimax guarantee leaves room for large improvements on particular reward distributions.

7.3 A Common Environment Across Random Arrival Scales

The third study holds the two reward means fixed before drawing the arrival schedule. At $T = 4^m$, $m = 4, 5, 6, 7$, the competing arm has lifetime $H = 4^J$, $J = 0, \ldots, m$. We consider probabilities proportional to 2^{-J}, uniform scale probabilities, probabilities proportional to 4^{-J}, an equal mixture of the last two scales, and a point mass at the last scale. The anchor mean is $1/2$; the competitor mean is $1/2 \pm \Delta$, with $\Delta = 2^{-1-r/2}$, $r = 0, \ldots, 2m + 2$.

One rule starts a fresh two-arm MOSS run with the remaining horizon revealed by the

known endpoint. The other is the original MC-MOSS, which does not know the endpoint and retains all anchor observations. Their panels specify this information difference. We simulate the forced prefix exactly through its binomial sufficient statistic.

For each policy and mixture, we use 64 development replications to select a single gap and sign, or a separate gap and sign for each lifetime. Both selections are then evaluated on 128 independent replications per lifetime. Mixture means use the known schedule probabilities, with stratified Monte Carlo standard errors. A separate check draws 2048 actual schedules for each selected common environment. These are held-out risks of development-selected members of a finite grid, not estimates of the full minimax risk or of a continuous supremum.

Figure 3 shows the conditional risk of a better arriving competitor. The difficult gap changes with the remaining lifetime. The retained prefix changes this surface, so the two panels also differ for reasons beyond endpoint information alone. Figure 4 then averages over the inverse-square-root mixture. At the largest endpoint, fresh-horizon MOSS has held-out mean regret 1.321 (standard error 0.018) under the selected common environment, compared with 1.977 (0.026) when the selected environment is allowed to depend on the schedule. The corresponding geometric values are $W(Q) = 1.412$ and $\mathbb{E}\sqrt{H} = 4.016$. The latter curve is an arrival-only reference, not an observed risk.

The remaining mixtures, direct schedule draws and selected gaps are reported in the computational supplement. For the two-long-scale mixture the held-out ordering is reversed: development selection can miss the best grid member. Independent evaluation removes reuse of selection noise from the reported means; it does not make finite-grid selection exact. None of these finite experiments estimates an asymptotic rate or establishes that the anytime policy attains the random-template theorem.

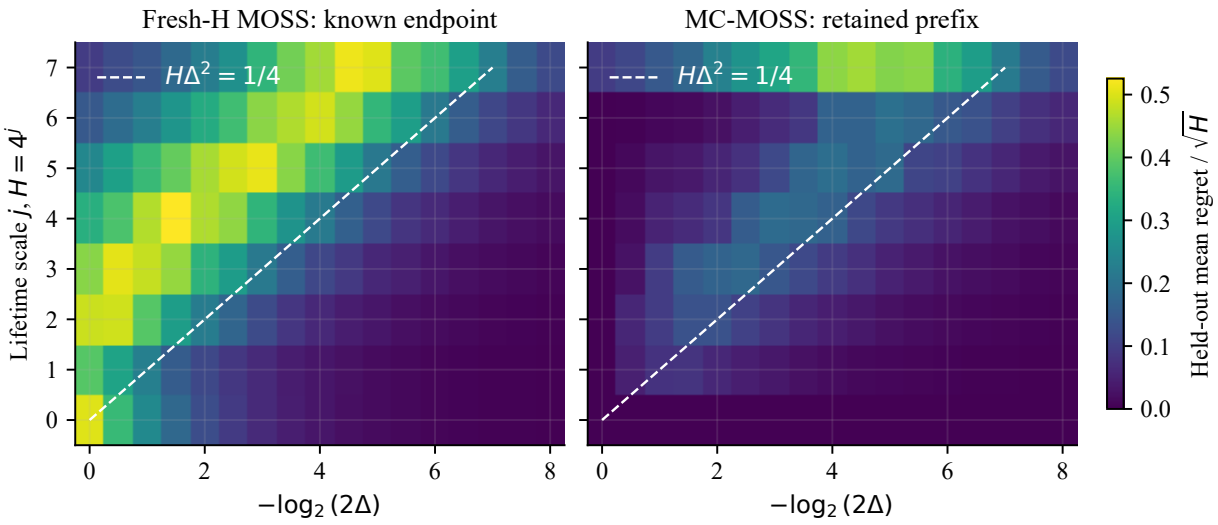


Figure 3. Held-out regret divided by $\sqrt{H}$ for a better arriving competitor at $T = 16384$. Each cell uses 128 replications. The dashed line marks $H\Delta^2 = 1/4$; it is a scale reference, not a fitted ridge. The policies differ in endpoint knowledge and retention of the forced anchor prefix.

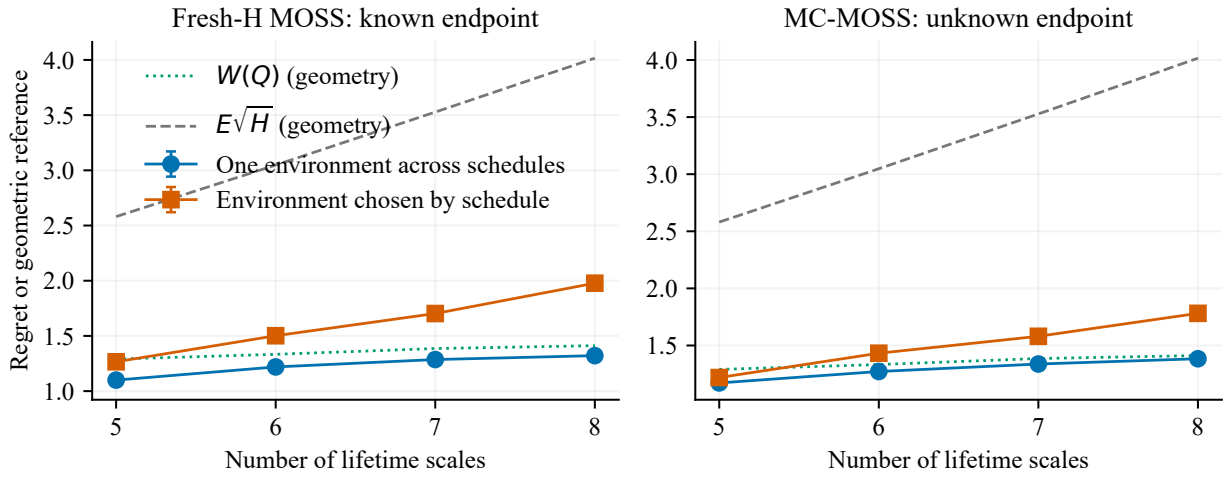


Figure 4. Common versus schedule-specific environments under $\Pr(J = j) \propto 2^{-j}$. Environments are selected on development replications and evaluated independently. Bars are 1.96 stratified Monte Carlo standard errors. The two geometric reference curves are computed from the arrival law and are not risk estimates. Selection is performed separately for each policy; the panels do not compare the policies under one common mean vector.

7.4 Two Fixed Batches Under a Random Common Stretch

We finally check the executable upper-bound construction on a template with two batches of two candidates, $\boldsymbol{\alpha} = (4, 1)$ and $K = 5$. Let $T = 4 \cdot 4^m$, $m = 3, 4, 5, 6$, and let H range over $1, 4, \ldots, 4^m$. We consider the inverse-square-root law $\mathbb{P}(H = 4^j) \propto 2^{-j}$, the uniform law on these scales, and the deterministic longest scale. Once the first batch arrives, the rule in Theorem 7 runs fresh fixed-horizon MOSS experiments of lengths $3H$ and H. The comparison is the original anytime MC-MOSS, which retains all observations, including the anchor's prefix. The former rule uses the known endpoint and revealed schedule; MC-MOSS is given neither.

The fixed reward vector is $(0.50, 0.45, 0.55, 0.55 + \eta\delta, 0.50)$, with $\eta \in \{-1, 1\}$ and $\delta \in \{0.025, 0.05, 0.10, 0.20, 0.40\}$. For each (m, Q) we use 64 development replications to select the single pair (η, δ) with the largest law-averaged regret for the revealed-stretch rule. We evaluate that same pair on 128 independent held-out replications for both policies and every value of H. Thus no environment is selected after the realized stretch. The finite law is averaged exactly over its support, with Monte Carlo sampling only for rewards. These are held-out risks for selected common environments, not estimates of the continuous minimax supremum.

At $m = 6$, the inverse-square-root law has $W(Q) = 1.386$; the revealed-stretch rule has mean regret 5.94 (standard error 0.05), and MC-MOSS has 3.11 (0.06). The corresponding revealed-stretch risks are 39.81 (0.58) under the uniform-scale law and 141.80 (3.14) under the deterministic longest scale, whose complexities are 12.190 and 64, respectively. Figure 5 displays the normalized values across all four endpoints. MC-MOSS performs better in every selected cell, despite using less schedule information. This does not contradict the theorem: the reset construction proves an upper bound and is not optimized for finite-sample constants.

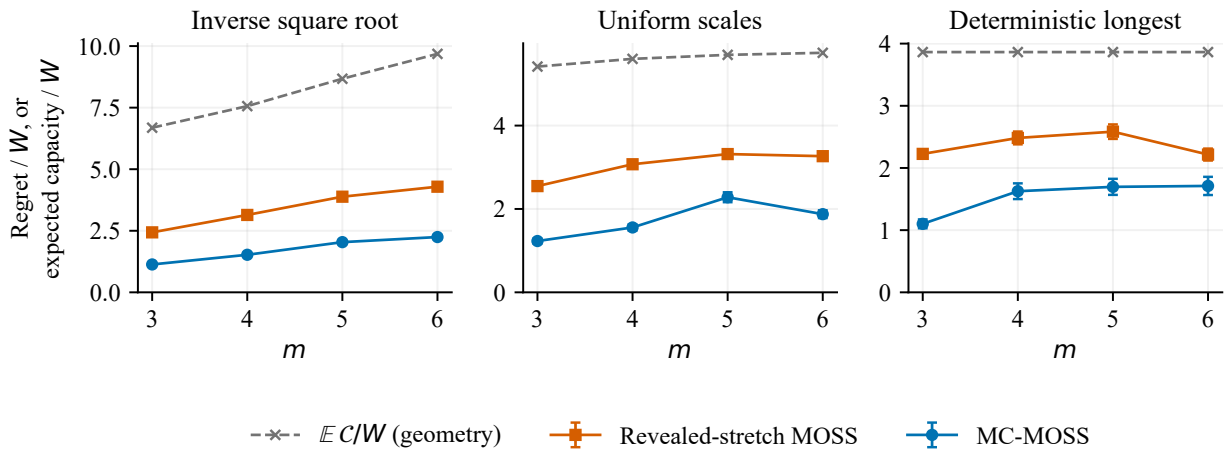


Figure 5. Held-out common-environment risks divided by $W(Q)$ for the fixed two-batch template. Error bars are 1.96 Monte Carlo standard errors under exact stratification over H. The dashed curve is the expected deterministic capacity divided by $W(Q)$ and is a geometric reference, not an observed regret. Both policies face the same development-selected environment in each panel and at each m.

Nor does this finite grid establish the asymptotic rate or a matching lower constant. Its role is to verify that the common-environment averaging and the multi-batch upper-bound rule can be implemented without discarding regret incurred before the last batch.

8. Discussion

The arrival schedule determines both which alternatives must be distinguished and how long their distinction can matter. For permanent arrivals with stationary bounded rewards, the schedule-specific minimax cost is an additive spectrum of remaining lifetimes. This result permits very late large batches to be inexpensive, while retaining the possibility that many separated scales each contribute to regret. An all-history causal policy achieves the resulting rate without estimating the future arrival process.

The allocation formulation gives a way to compare schedules before observing rewards. It can distinguish arrival plans with identical endpoints and arm counts, quantify the effect of timing perturbations, and express the marginal contribution of adding candidates. These are comparisons of worst-case learning difficulty. They do not measure the expected benefit of an arm whose mean is known or believed to be high. Using the capacity to design an admission policy would require a model for those benefits and a revised benchmark.

The mechanism experiments qualify how the optimal rate should guide implementation. Clock changes and capacity updates can have opposing finite-sample effects, so a comparison that changes both cannot identify either effect alone. Retaining observations avoids repeated exploration in some environments. The boundary examples and baseline comparisons show that the same minimax rate can conceal substantially different exploration costs near the reward upper bound. The gap bound depends on the smallest positive gaps and may be loose for heterogeneous instances. A general arrival-aware instance complexity, together with a policy attaining it, remains to be established.

The probability lower bound in the supplement shows that spectral difficulty is present with constant probability. The accompanying upper probability bounds come from expectation and are not sharp for small error probabilities. Likewise, the deterministic-time expectation bound alone makes no assertion at reward-dependent stopping times. Reward-dependent arrivals would also invalidate the simple conditioning and deterministic comparator-label arguments used here, and require a separate analysis.

Independent random arrivals change which hard environments can coexist. Theorem 7 resolves this issue for fixed, revealing batch templates: a truncated-lifetime functional replaces the average conditional capacity, and the ratio $D(Q)$ states when averaging is adequate. The result depends on a known endpoint and a fixed template, and its attaining policy uses the schedule information revealed by the first batch. A characterization for nonrevealing random

schedules or for a growing collection of batches would have to account for information about future arrivals as well as information about reward means. The simulations illustrate the common-environment restriction without claiming to estimate that more general minimax problem.

Supplementary Material

The mathematical supplement provides all proofs and additional probability and gap-dependent results. A separate computational supplement gives complete simulation designs, sensitivity analyses, baseline comparisons and the earlier fixed-endpoint study. The reproducibility package contains the policies, complexity calculations, simulation seeds, per-replicate results and scripts for every reported table and figure.

Data and Code Availability

All numerical data in this article are generated by the accompanying Python scripts. The reconstruction package includes the complete source, environment versions, a master execution script and a mapping from scripts to reported outputs. It uses no restricted or external individual-level data.

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